

Jonathan Liu

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EDUCATION

University of Washington

Bachelor of Science in Electrical and Computer Engineering | GPA: 3.93

Seattle, WA

Sep 2024 – June 2027

- **Honors:** IEEE HKN Honor Society (Top 8% ECE), Tau Beta Pi Engineering Honor Society (Top 12.5%).
- **Relevant Coursework:** Analog IC Design, Microelectronic Circuits, Digital Circuit Design, Signals & Systems, Power Electronics, Control Systems, Computer Architecture, Data Structures, Hardware/Software Interface.

Tohoku University STEM Summer Program

Nanoelectronics Fabrication & Robotics Immersion | GPA: 4.0

Sendai, Japan

June – July 2025

SKILLS

VLSI: Cadence Virtuoso (Schematic & Layout), ADE Explorer, ADE Assembler, GPDK045 (45nm CMOS), LTspice, SPICE, Op-Amps, OTAs, Current Mirrors, Differential Pairs, Frequency Compensation.

Hardware & Embedded: Altium Designer, KiCad, STM32 (CubeIDE/CubeMX/HAL), Arduino, FPGA (SystemVerilog), Intel Quartus, ModelSim, BLDC Motor Control, LDOs, Soldering, Oscilloscopes, Logic Analyzers.

Software & Simulation: C/C++, Python (NumPy/SciPy/PyTorch/scikit-learn), Java, MATLAB, Fusion360, OpenSCAD, Linux/Bash, Git.

Languages: English, Chinese (Mandarin & Cantonese), Japanese.

RELEVANT EXPERIENCE

Undergraduate Research Assistant

Sensors, Energy, and Automation Laboratory (SEAL)

Jan 2026 – Present

Seattle, WA

- Co-investigating a cross-modality study validating **PPG-derived heart-rate variability** against synchronous **ECG** across multiple body sites and Fitzpatrick skin types; manuscript in preparation for *MDPI Sensors*.
- Implementing a multi-feature **HRV analytics pipeline** in **Python**, with Lin's CCC and Bland-Altman concordance analysis against ECG reference.
- Built the multi-site acquisition stack: **MAX30102** PPG and **AD8232** ECG sensors with custom **3D-printed** body-sensor mounts and a live FastAPI dashboard.

Lead (Sep 2025 – Present) | Drone Hardware Engineer

Husky Robotics Drone Subteam

Oct 2024 – Present

Seattle, WA

- **Technical Lead:** Spearhead recruitment and technical onboarding for 20+ members; manage the full development lifecycle of three custom **PCBs** (Flight Controller, ESC, PDB), BOM selection, and Professional Design Reviews.
- **Drone System Architecture:** Architected and built a competition-grade autonomous quadcopter from spec to flight test for the URC 2026 Delivery mission, integrating propulsion, **6S Li-Ion** power, **ExpressLRS** RC, and a **Jetson Orin Nano** autonomy stack with **MAVLink**-over-WiFi telemetry.
- **Custom Flight Controller PCB:** Architected a Pixhawk-class flight controller from scratch in **Altium Designer** around an **STM32H743VGT6**; completed full schematic and **4-layer PCB layout** with integrated IMU, barometer, current/power sensing, and USB-C power delivery.
- **Embedded Firmware:** Developed bare-metal **STM32 HAL** drivers in **CubeIDE/CubeMX** for **DSHOT600** motor control, **MAVLink** telemetry, and **PX4**-compatible sensor fusion.

Autonomous Systems Engineering Intern

ARIDGE (XPeng AEROHT)

July – Sep 2025

Guangzhou, China

- Trained **computer vision** models for autonomous landing-zone classification and validated **LiDAR** sensor-fusion integrity on the XPeng HT Aero X3 flying car platform.

PROJECTS

5T-OTA-Based CMOS Operational Amplifier (GPDK045)

Spring 2026

- Designing a single-stage **5T OTA** in **Cadence Virtuoso** (**GPDK045** 45nm CMOS) as a closed-loop inverting op-amp; hand-sizing devices from first-principles small-signal analysis and validating in **ADE Explorer**.

Adjustable AC-DC Boost Converter

May 2026

- Designed an adjustable **AC-DC boost converter** in **LTspice** (± 7.5 VAC $\rightarrow$ +10–20 VDC) using an NE555-clocked 2N7000 boost stage with op-amp/Zener feedback regulation; breadboard validation in progress.

Three-Band Active Audio Equalizer

March 2026

- Designed and constructed a three-band active audio equalizer around the **LM348** op-amp; derived transfer functions and validated frequency response via full-system **LTspice** simulation.

FPGARhythm

March 2026

- Implemented a hardware rhythm game in **SystemVerilog** on a **Cyclone V GX** (DE1-SoC) featuring PLL-derived **VGA** timing, LFSR-driven note sequencing, debounced input handling, and **I²S** audio playback via a WM8731 CODEC.

ORGANIZATIONS & AWARDS

- **Dean's List** – UW (2024–Present); **UW Line-Following Car** – All-Time Fastest Record (Feb 2026); **Events Officer** – IEEE UW & HKSA at UW.